

SAMUEL DIBABU ASSEFA

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EDUCATION

Tulane University Celia Scott Weatherhead School of Public Health and Tropical Medicine New Orleans, LA
Master of Science in Biostatistics and Data Science (Concentration in Statistical Machine Learning), GPA 4.0 May 2027

- Honors: 1912 Founders Scholar
- Relevant Coursework: Statistical Inference I&II, Statistical Learning in Data Science, Introduction to methods in Data Science (Unix/Linux Operating Systems, Git Bash, Python, R studio, SQL), Data Management and Statistical Analysis, Database Management, Intermediate Biostatistics, Categorical Data Analysis, Design of Experiments, Regression Analysis, Epidemiology

Addis Ababa University

Addis Ababa, Ethiopia

Bachelor of Science in Diagnostic Medical Imaging, GPA 3.86

July 2022

- Honors: First Class Honors
- Relevant Coursework: Radiologic Physics, Radiologic Equipment, Radiologic Anatomy, Radiologic Pathology, Medical Ultrasonography, Computed Tomography and Magnetic Resonance Imaging

WORK EXPERIENCE

Tulane University Department of Biostatistics and Data Science | Kakraba Research Lab

New Orleans, LA

Graduate Research Assistant

June 2026 – Present

- Engineer and benchmark comprehensive machine learning pipelines in Python and R implementing linear and semi-parametric modeling (Logistic Regression, GAM), discriminant analysis (LDA, QDA), probabilistic methods (Naive Bayes), instance and kernel algorithms (KNN, SVC), tree-based ensembles (Random Forest, Extra Trees), advanced boosting frameworks (AdaBoost, XGBoost, LightGBM, CatBoost), and unsupervised architectures (PCA, clustering) to optimize risk prediction in clinical medicine and public health context

Tulane University Violence Prevention Institute

New Orleans, LA

Graduate Research Assistant

October 2025 – Present

- Conduct data cleaning, feature engineering, geospatial mapping, descriptive and inferential statistical data analysis for longitudinal studies conducted on gun-violence as part of the E2E and SHIFT projects.

Tulane University Department of Biostatistics and Data Science

New Orleans, LA

Graduate Teaching Assistant - SPHL 6050 (Biostatistics for Public Health)

August 2026 – Present

- Guide public health graduate students on introductory biostatistical concepts and statistical analysis using SPSS

PROJECTS

Explainable AI–Driven Survival Prediction and Risk Stratification in Heart Failure

Python | Scikit-Learn | LightGBM | XGBoost | SHAP | Git

- Trained nine machine learning classification models on a clinical cohort of 299 heart failure patients from the UCI repository to optimize risk-stratification accuracy.
- Implemented post-hoc Explainable AI (XAI) using global and local SHAP value frameworks grounded in cooperative game theory to map non-linear risk attributions of key clinical features

Chronic Disease Surveillance Pipeline for Orleans Parish

R | SQL | SQLite | Bash | Unix | GitHub | Power BI

- Automated data cleaning, SQL cohort analysis, and statistical summaries from raw health datasets.
- Integrated Unix data processing, database querying, and R scripting into a single executable workflow.
- Delivered population health insights through automated reports and visualization dashboards.

Orleans Parish Socioeconomic Trend Visualization Shiny-app

R | Shiny | tidycensus | ggplot2 | Git

- Constructed a longitudinal census tract–level panel dataset (2016–2023) using the U.S. Census ACS 5-year estimates via the Census API.
- Conducted feature engineering on demographic and socioeconomic indicators, computed means across 2016–2023.
- Designed and deployed an interactive R Shiny dashboard to visualize longitudinal socioeconomic trends (2016–2023), with a publicly accessible app and fully reproducible codebase on GitHub.

PUBLICATIONS

- *“Explainable Machine Learning for Heart Failure Mortality Prediction: Comparative Modeling, Leakage-Aware Validation, and Independent Cardiovascular Benchmarking”* MedComm [Under Review]

- “Using Supervised and Unsupervised Learning to Predict Life Expectancy and Examine Socioeconomic/Health Predictors” [Pending]
- “From Cognitive Sovereignty to AI Economic Sovereignty: A Phased Implementation Roadmap for Equitable AI in Global Health” [Pending]
- “Assessing Community-Level Impact of a Youth-Led Sociopolitical Action Campaign to Prevent Community and Police Violence in New Orleans” [Pending]
- “Preventing Youth Firearm Violence in New Orleans: A Hospital-Initiated Community Integrated Approach” [Pending]

PRESENTATIONS

Annual Tulane Research, Innovation, and Creativity Summit (TRICS 2026)

April 2026

[Machine Learning–Driven Survival Prediction and Risk Stratification in Heart Failure](#)

AWARDS

2nd place winner | SEER Hackathon | Tulane Data Analytics and Science Hub

October 2025

[Socioeconomic and Rurality Effects on Mortality for Early-Onset Colorectal Cancer: A SEER Population-based Analysis](#)

SKILLS AND CERTIFICATIONS

Languages: English (Full Professional), Amharic (Native)

Technical Skills: Python Ecosystem (PyTorch, Scikit-Learn, LightGBM, XGBoost, SHAP, LIME, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter Notebook), Statistical Computing in R (Tidyverse, dplyr, ggplot2, Survival, Survminer, Lme4, Tidymodels, Shapr, R Shiny, Tidycensus, R Markdown), Linux/Unix Operating Systems, Bash Shell Scripting, Git Version Control System, SQL,

Soft Skills: Problem Solving, Analytic Thinking, Technical Communication, Leadership, Cross-functional Collaboration

Certifications: CITI IRB Responsible Conduct of Research (RCR) – Biomedical & Social and Behavioral

Louisiana Clinical and Translational Science Center – Level I, II & III Clinical Research Coordinator

EXTRA-CURRICULAR ACTIVITY

Event Assistant | LA CaTS Annual Coordinator Training | Tulane Center for Clinical Research January 2026 – Present